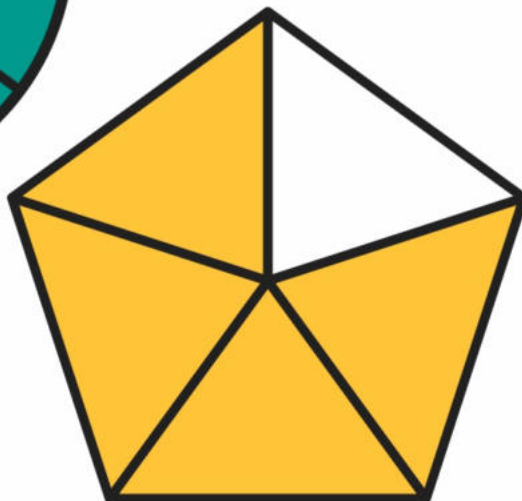
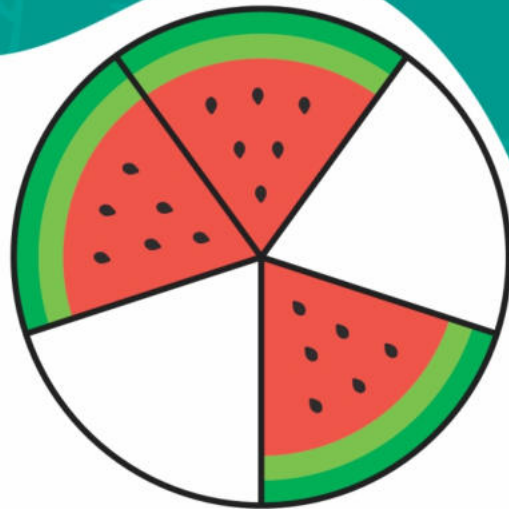


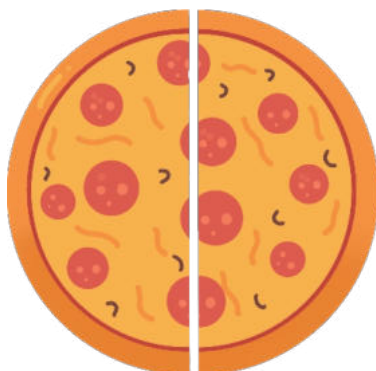
5

FRACTIONS



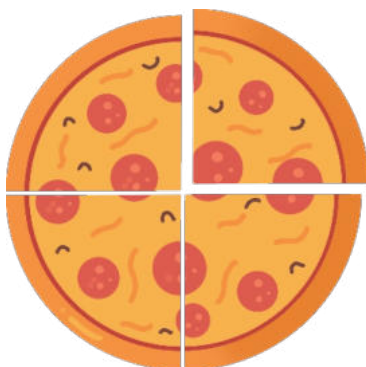
● Definition

- A fraction represents a part of a whole. It tells us how many parts of a certain size we have.



For example:

A pizza divided into 4 equal slices can show $\frac{1}{4}$, $\frac{1}{2}$ etc.



● Parts of a Fraction

A fraction has two main parts:

- **Numerator:** The top number that shows how many parts we have.

3. — Numerator

— • — Fraction bar

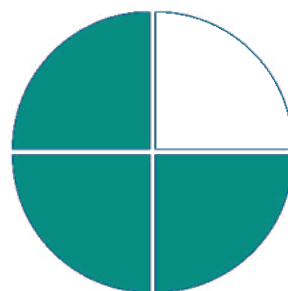
4. — Denominator

- **Denominator:** The bottom number that shows how many parts make up the whole.

For example:

For example, in the fraction $\frac{3}{4}$:

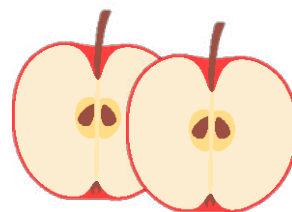
- **3** is the numerator.
- **4** is the denominator.



● Examples of Fractions

- $\frac{1}{2}$: Represents half of something.
- $\frac{1}{3}$: Represents one-third of something.
- $\frac{3}{4}$: Represents three out of four equal parts.

Fractions help us share things equally. For example, if you have an apple and cut it into 2 equal parts, each part is $\frac{1}{2}$ of the apple.



A chocolate bar broken into 2 pieces $\frac{1}{2}$.

chocolate bar

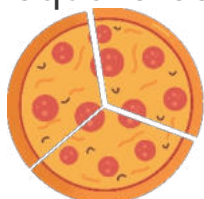


$\frac{1}{2}$ bar each



An pizza divided into 3 equal sections $\frac{1}{3}$.

3 equal slices



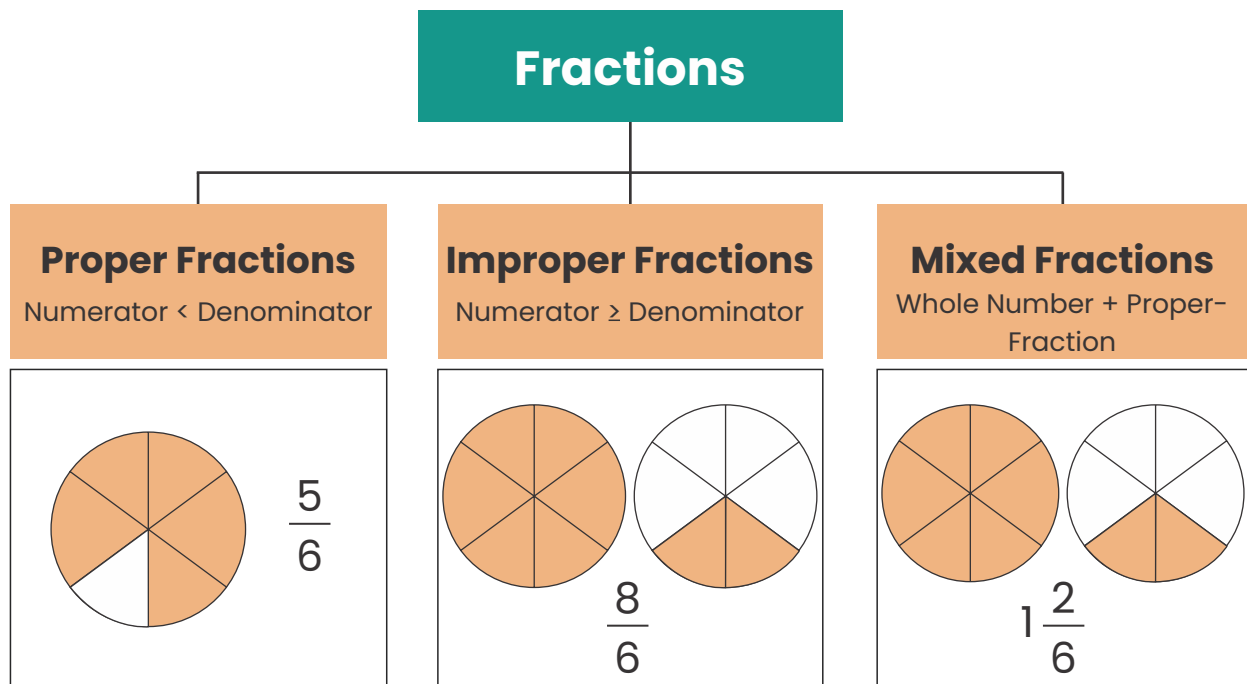
Each slice = $\frac{1}{3}$

A pizza with 3 out of 4 parts shaded $\frac{3}{4}$.





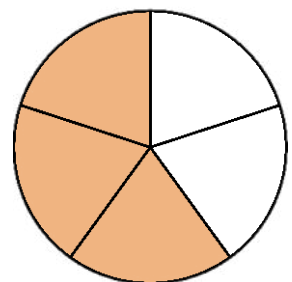
Types of Fractions



Proper Fractions

The numerator is smaller than the denominator.

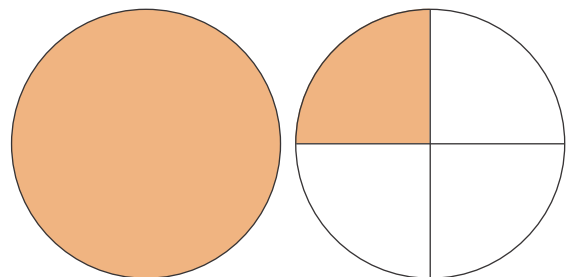
Example: $\frac{3}{5}$



Improper Fractions

The numerator is greater than or equal to the denominator.

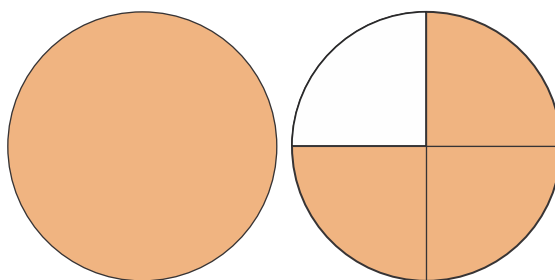
Example: $\frac{5}{4}$



● Mixed Numbers

A mixed number has a whole number and a fraction.

Example: $1\frac{1}{2}$ (One and a half).

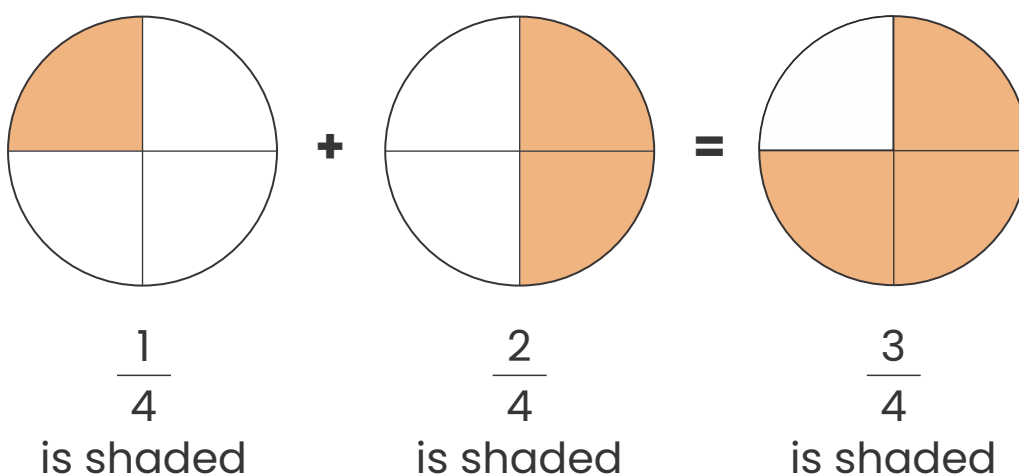


$1\frac{3}{4}$ Circle

● Like Fractions

Like fractions have the same denominator.

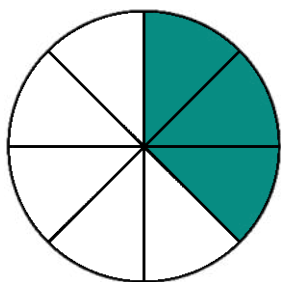
Example: $\frac{1}{4}$ and $\frac{3}{4}$ are like fractions.



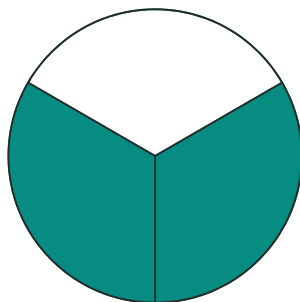
• Unlike Fractions

Unlike fractions have different denominators.

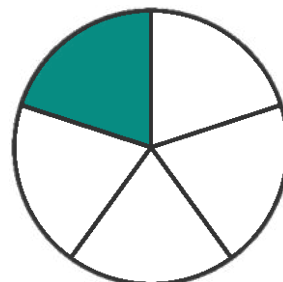
Example: $\frac{1}{3}$ and $\frac{1}{5}$ are unlike fractions.



$$\frac{3}{8}$$



$$\frac{1}{3}$$

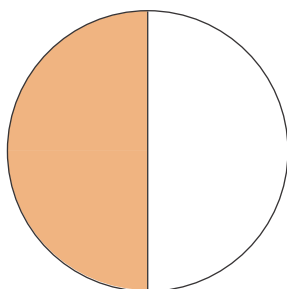


$$\frac{1}{5}$$

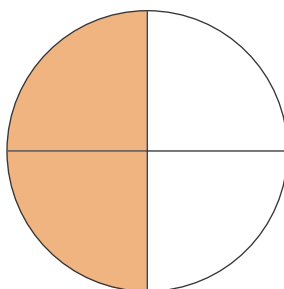
• Equivalent Fractions

Equivalent fractions are different fractions that represent the same value.

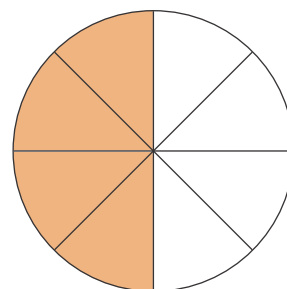
Example: $\frac{1}{2}$ is the same as $\frac{2}{4}$.



$$\frac{1}{2}$$



$$\frac{2}{4}$$



$$\frac{4}{8}$$

● Identify the fractions as Proper. Improper or Mixed.

1) $\frac{4}{7} =$

2) $\frac{3}{7} =$

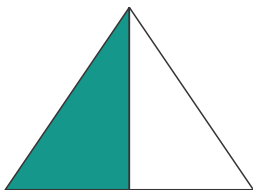
3) $3\frac{6}{11} =$

4) $\frac{9}{4} =$

5) $3\frac{1}{5} =$

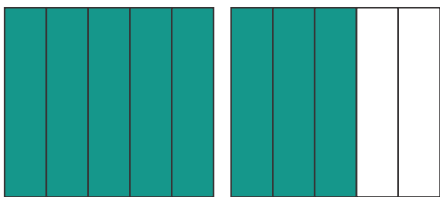
● Write the fraction.

1) $\frac{\boxed{1}}{\boxed{}} =$



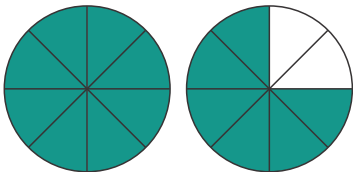
Unit Fractions

2) $\frac{\boxed{}}{\boxed{5}} =$



Mixed Fractions

3) $\boxed{} \frac{\boxed{}}{\boxed{8}} =$



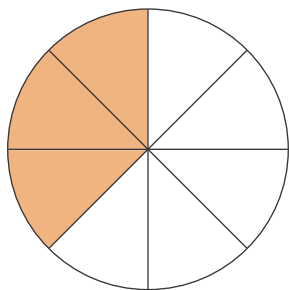
Proper Fractions

• Comparing and Ordering Fractions

Fractions show parts of a whole. We can compare them to see which fraction is larger or smaller.

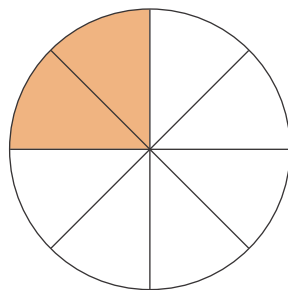
Example: Which is bigger: $\frac{1}{4}$ or $\frac{1}{2}$?

Answer: $\frac{1}{2}$ is bigger because it represents a larger part of the whole.



$$\frac{3}{8}$$

>



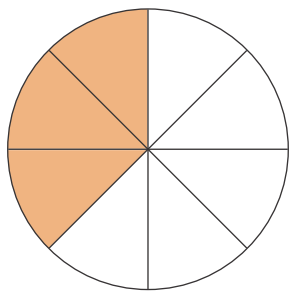
$$\frac{2}{8}$$

denominators
are the same,

8

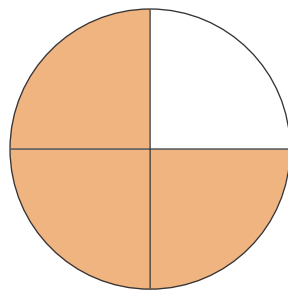
so compare
numerators:

$$3 > 2$$



$$\frac{3}{8}$$

<



$$\frac{3}{4}$$

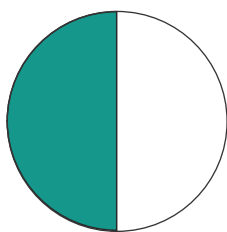
numerators
are the same,

3

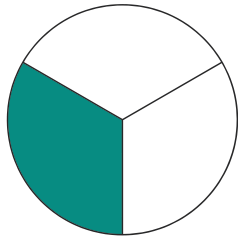
so compare
which slice
is bigger:

$$\frac{1}{8} < \frac{1}{4}$$

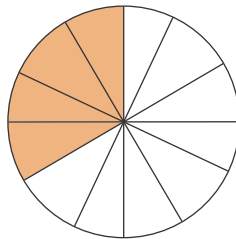
- Compare the fractions, and write $>$ or $<$ or $=$ between them.



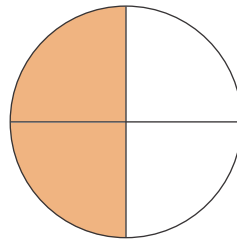
$$\frac{1}{2}$$



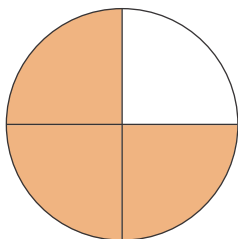
$$\frac{1}{3}$$



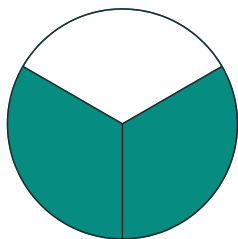
$$\frac{4}{12}$$



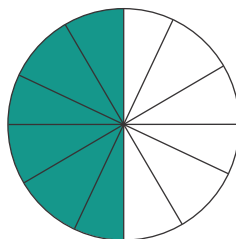
$$\frac{2}{4}$$



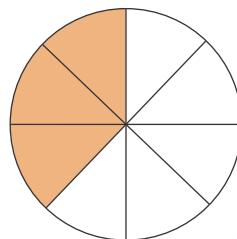
$$\frac{3}{4}$$



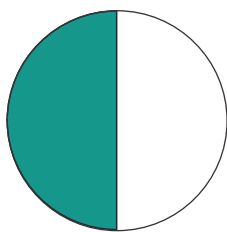
$$\frac{2}{3}$$



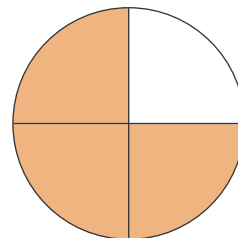
$$\frac{6}{12}$$



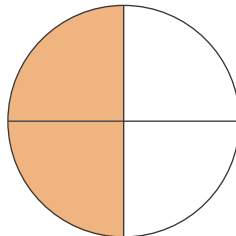
$$\frac{3}{8}$$



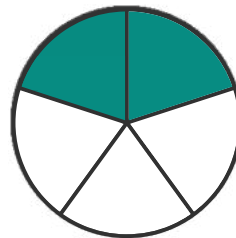
$$\frac{1}{2}$$



$$\frac{3}{4}$$



$$\frac{2}{4}$$



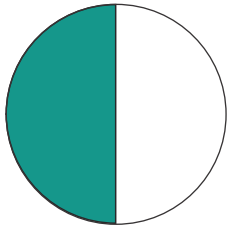
$$\frac{2}{5}$$



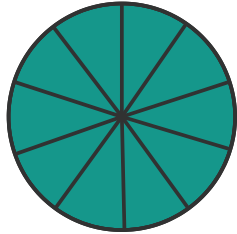
FRACTIONS



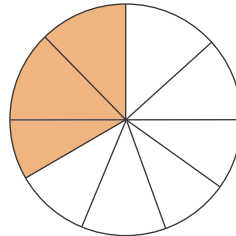
- Compare the fractions, and write $>$ or $<$ or $=$ between them. You can colour the pieces in the pie pictures to help you.



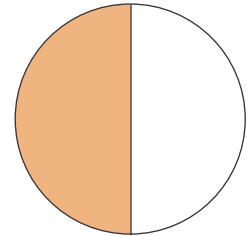
$$\frac{1}{2}$$



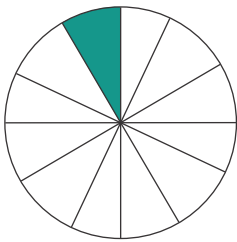
$$\frac{10}{10}$$



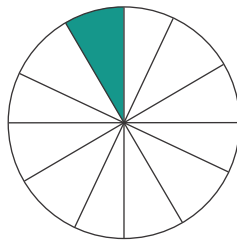
$$\frac{3}{9}$$



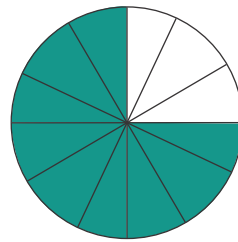
$$\frac{1}{12}$$



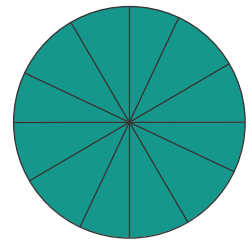
$$\frac{1}{12}$$



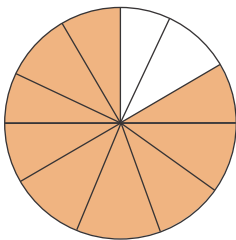
$$\frac{1}{12}$$



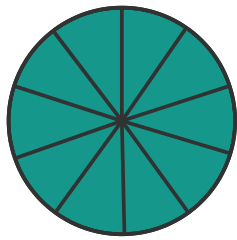
$$\frac{9}{12}$$



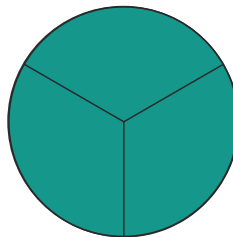
$$\frac{12}{12}$$



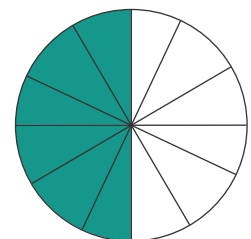
$$\frac{9}{11}$$



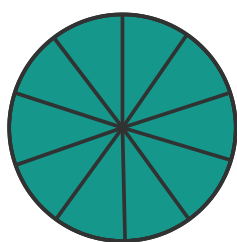
$$\frac{10}{10}$$



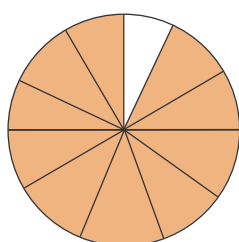
$$\frac{3}{3}$$



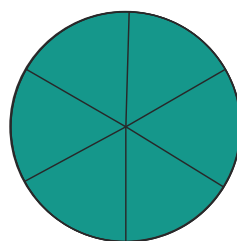
$$\frac{6}{12}$$



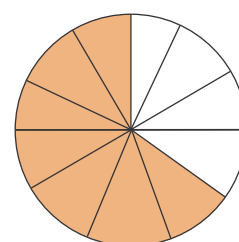
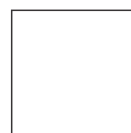
$$\frac{10}{10}$$



$$\frac{10}{11}$$



$$\frac{6}{6}$$



$$\frac{7}{11}$$

● Converting Improper Fractions into Mixed Numbers.

An improper fraction has a numerator that is larger than or equal to the denominator. To convert an improper fraction to a mixed number, divide the numerator by the denominator. The quotient is the whole number, and the remainder becomes the numerator of the fraction.

● Example: Convert $\frac{9}{4}$ into a mixed number.

● **Step 1:** Divide 9 by 4. The quotient is 2, and the remainder is 1.

● **Step 2:** Write the quotient as the whole number and place the remainder over the original denominator.

$$\text{So, } \frac{9}{4} = 2 \frac{1}{4}$$

● Converting Mixed Numbers into Improper Fractions.

● Example: Convert $1 \frac{1}{2}$ into an improper fraction.

● **1.** Multiply the whole number (1) by the denominator (2):
 $1 \times 2 = 2$



2. Add the numerator (1) to this result:

$$2 + 1 = 3$$

3. Place this result over the original denominator:

The improper fraction is $\frac{3}{2}$

So, $1\frac{1}{2}$ is equal to $\frac{3}{2}$.

- **Convert the following mixed numbers to improper fractions:**

1) $2\frac{2}{3} =$

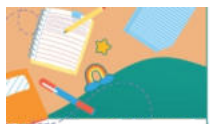
2) $3\frac{3}{4} =$

3) $3\frac{1}{4} =$

4) $1\frac{1}{2} =$

5) $4\frac{2}{5} =$

6) $3\frac{4}{5} =$



FRACTIONS



7) $2 \frac{3}{5} =$

8) $2 \frac{5}{6} =$

9) $1 \frac{1}{3} =$

10) $1 \frac{1}{2} =$

- Convert each improper fraction to a mixed number.

1) $\frac{11}{7} =$

2) $\frac{5}{3} =$

3) $\frac{9}{5} =$



FRACTIONS



4) $\frac{9}{8} =$

5) $\frac{17}{12} =$

6) $\frac{10}{4} =$

7) $\frac{18}{11} =$

8) $\frac{7}{5} =$

9) $\frac{10}{5} =$

10) $\frac{5}{4} =$

11) $\frac{7}{6} =$

12) $\frac{22}{15} =$



● Equivalent Fractions

Equivalent fractions are different fractions that represent the same part of a whole. For example, $\frac{1}{2}$ is equivalent to $\frac{2}{4}$, $\frac{3}{6}$, and $\frac{4}{8}$.

To find equivalent fractions, multiply or divide both the numerator and the denominator by the same number.

- **Example:** Multiply the numerator and denominator of $\frac{1}{3}$ by 2 to get $\frac{2}{6}$, which is equivalent to $\frac{1}{3}$.

● Visualizing Equivalent Fractions

Equivalent fractions represent the same part of a whole even though they look different. For instance, $\frac{1}{3}$ is equivalent to $\frac{2}{6}$ and $\frac{4}{12}$.

- **1/3:** Circle divided into **3** parts, **1** part shaded.
- **2/6:** Circle divided into **6** parts, **2** parts shaded.
- **4/12:** Circle divided into **12** parts, **4** parts shaded.

● Fill in the Blanks to Make Equivalent Fractions.

$$1) \frac{11}{7} = \frac{\quad}{14}$$

$$2) \frac{2}{6} = \frac{\quad}{3}$$

$$3) \frac{4}{12} = \frac{\quad}{3}$$

$$4) \frac{1}{3} = \frac{\quad}{12}$$

$$5) \frac{2}{6} = \frac{\quad}{12}$$

$$6) \frac{4}{12} = \frac{\quad}{6}$$



FRACTIONS



- Complete the equivalent fractions by filling in the blanks:

$$1) \frac{5}{10} = \frac{\quad}{90}$$

$$2) \frac{2}{4} = \frac{\quad}{20}$$

$$3) \frac{3}{7} = \frac{\quad}{28}$$

$$4) \frac{15}{24} = \frac{\quad}{3}$$

$$5) \frac{9}{2} = \frac{\quad}{40}$$

$$6) \frac{6}{15} = \frac{\quad}{45}$$

$$7) \frac{6}{27} = \frac{\quad}{9}$$

$$8) \frac{39}{12} = \frac{\quad}{4}$$

$$9) \frac{10}{3} = \frac{\quad}{27}$$

$$10) \frac{1}{6} = \frac{\quad}{54}$$

$$11) \frac{4}{4} = \frac{\quad}{12}$$

$$12) \frac{1}{2} = \frac{\quad}{8}$$

$$13) \frac{4}{12} = \frac{\quad}{6}$$

$$14) \frac{1}{4} = \frac{\quad}{48}$$

$$15) \frac{1}{7} = \frac{\quad}{28}$$



● Addition of Like Fractions

When adding fractions with the same denominators, simply add the numerators and keep the same denominator.

For example:

$$\frac{2}{7} + \frac{3}{7} = \frac{2+3}{7} = \frac{5}{7}$$

● Add the following like fractions.

1) $\frac{2}{7} + \frac{3}{7} =$

2) $\frac{5}{9} + \frac{1}{9} =$

3) $\frac{4}{8} + \frac{2}{8} =$

4) $\frac{1}{6} + \frac{4}{6} =$

5) $\frac{7}{10} + \frac{2}{10} =$

● Addition of Unlike Fractions

When adding fractions with **different denominators**, follow these steps:

1. Find the Least Common Denominator (LCM) of the denominators.

2. Convert each fraction to an equivalent fraction with the common denominator.

3. Add the numerators and keep the common denominator.

For example:

$$\frac{3}{4} + \frac{2}{5}$$

1. The least common denominator of 4 and 5 is 20.

2. Convert the fractions:

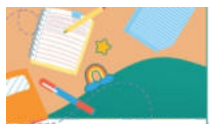
$$\frac{3}{4} = \frac{15}{20} \text{ (Multiply the numerator and denominator by 5)}$$

$$\frac{2}{5} = \frac{8}{20} \text{ (Multiply the numerator and denominator by 4)}$$

3. Add the numerators:

$$\frac{15}{20} + \frac{8}{20} = \frac{23}{20} = 1 \frac{3}{20}$$

$$\text{So, } \frac{3}{4} + \frac{2}{5} = 1 \frac{3}{20}$$



FRACTIONS



- Add the following like fractions.

1) $\frac{1}{2} + \frac{1}{3} =$

2) $\frac{5}{6} + \frac{2}{5} =$

3) $\frac{3}{7} + \frac{3}{5} =$

4) $\frac{5}{8} + \frac{2}{3} =$

5) $\frac{1}{3} + \frac{3}{5} =$

6) $\frac{2}{5} + \frac{5}{6} =$

7) $\frac{1}{4} + \frac{1}{6} =$

8) $\frac{2}{5} + \frac{9}{8} =$

9) $\frac{1}{6} + \frac{3}{4} =$



FRACTIONS



- Solve the following problems using the steps for adding unlike fractions.

1) $\frac{1}{4} + \frac{2}{12} =$

2) $\frac{2}{6} + \frac{1}{2} =$

3) $\frac{1}{3} + \frac{2}{9} =$

4) $\frac{5}{8} + \frac{1}{16} =$

5) $\frac{2}{9} + \frac{1}{27} =$

6) $\frac{1}{3} + \frac{3}{21} =$

7) $\frac{3}{4} + \frac{1}{16} =$

8) $\frac{1}{5} + \frac{1}{10} =$

9) $\frac{3}{7} + \frac{1}{28} =$



• Subtraction of Like Fractions

When subtracting fractions with the same denominator, simply subtract the numerators and keep the denominator the same.

For example:

$$\frac{5}{8} - \frac{3}{8}$$

Since the denominators are the same, subtract the numerators:

$$\frac{5-3}{8} = \frac{2}{8}$$

• Subtract the like fractions.

1) $\frac{8}{2} - \frac{1}{2} =$

2) $\frac{2}{7} - \frac{1}{7} =$

3) $\frac{4}{6} - \frac{3}{6} =$

4) $\frac{5}{6} - \frac{4}{6} =$

5) $\frac{5}{6} - \frac{3}{6} =$



$$6) \frac{6}{7} - \frac{2}{7} =$$

$$7) \frac{3}{3} - \frac{1}{3} =$$

$$8) \frac{4}{4} - \frac{1}{4} =$$

$$9) \frac{5}{5} - \frac{1}{5} =$$

$$10) \frac{4}{4} - \frac{2}{4} =$$

● Subtraction of Unlike Fractions

When subtracting fractions with **different denominators**, follow these steps:

1. Find the Least Common Denominator (LCM) of the denominators.

2. Convert each fraction to an equivalent fraction with the common denominator.

3. Subtract the numerators and keep the common denominator.



For example:

$$\frac{7}{10} - \frac{2}{5}$$

1. Find a common denominator (LCM of 10 and 5 is 10).

2. Convert $\frac{2}{5}$ to $\frac{4}{10}$ by multiplying the numerator and denominator by 2.

3. Subtract the fractions:

$$= \frac{7}{10} - \frac{4}{10}$$

$$= \frac{3}{10}$$

• **Subtract the unlike fractions.**

1) $\frac{4}{8} - \frac{2}{5} =$

2) $\frac{4}{5} - \frac{1}{7} =$

3) $\frac{2}{7} - \frac{1}{5} =$

4) $\frac{9}{8} - \frac{2}{2} =$

5) $\frac{2}{2} - \frac{3}{5} =$



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$$6) \frac{5}{7} - \frac{2}{4} =$$

$$7) \frac{7}{8} - \frac{2}{6} =$$

$$8) \frac{4}{6} - \frac{2}{7} =$$

- Solve the following addition and subtraction problems.

$$1) \frac{1}{4} + \frac{2}{12} =$$

$$2) \frac{1}{3} + \frac{3}{21} =$$

$$3) \frac{2}{6} + \frac{1}{2} =$$

$$4) \frac{2}{2} + \frac{3}{5} =$$

$$5) \frac{1}{5} + \frac{1}{10} =$$

$$6) \frac{5}{8} + \frac{1}{16} =$$



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7) $\frac{3}{7} + \frac{1}{18} =$

8) $\frac{2}{9} + \frac{1}{27} =$

9) $\frac{3}{5} + \frac{1}{10} =$

10) $\frac{3}{4} + \frac{1}{16} =$

11) $\frac{1}{2} - \frac{1}{4} =$

12) $\frac{5}{6} - \frac{1}{3} =$

13) $\frac{3}{4} - \frac{1}{8} =$

14) $\frac{7}{9} - \frac{2}{9} =$

15) $\frac{4}{5} - \frac{3}{10} =$



$$16) \frac{2}{7} - \frac{1}{14} =$$

$$17) \frac{9}{16} - \frac{5}{16} =$$

$$18) \frac{11}{12} - \frac{7}{12} =$$

$$19) \frac{3}{5} - \frac{2}{15} =$$

$$20) \frac{7}{8} - \frac{1}{4} =$$

● Multiplying Fractions

When we multiply fractions, we multiply **top × top** and **bottom × bottom**.

Top number = numerator.

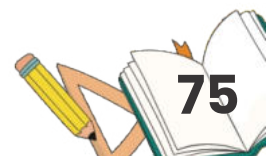
Bottom number = denominator

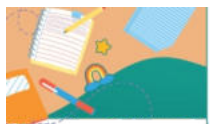
How to do it:

Multiply the numerators. Multiply the denominators. Write the new fraction. Simplify if possible.

Example:

$$\frac{2}{5} \times \frac{3}{7} = \frac{2 \times 3}{5 \times 7} = \frac{6}{35}$$





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- **Dividing Fractions**

To divide fractions, we flip the second fraction and then multiply. Flipping a fraction gives its reciprocal (example: $\frac{3}{4}$ becomes $\frac{4}{3}$).

How to do it:

Keep the first fraction the same. Flip the second fraction. Change $\div$ to $\times$. Multiply normally. Simplify if possible.

Example:

$$\frac{1}{2} \div \frac{3}{4} = \frac{1}{2} \times \frac{4}{3} = \frac{1 \times 4}{2 \times 3} = \frac{4}{6} = \frac{2}{3}$$

- **Solve the multiplication**

1) $\frac{2}{4} \times \frac{3}{4} =$

2) $\frac{2}{5} \times \frac{2}{5} =$

3) $\frac{1}{3} \times \frac{2}{2} =$

4) $\frac{3}{6} \times \frac{3}{6} =$

5) $\frac{7}{8} \times \frac{2}{8} =$



6) $\frac{5}{8} \times \frac{7}{8} =$

7) $\frac{1}{5} \times \frac{3}{5} =$

8) $\frac{4}{5} \times \frac{1}{5} =$

9) $\frac{2}{3} \times \frac{1}{3} =$

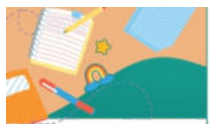
10) $\frac{2}{8} \times \frac{5}{8} =$

• Solve the multiplication.

1) $3\frac{2}{8} \times \frac{9}{10} =$

2) $1\frac{1}{3} \times \frac{3}{8} =$

3) $2\frac{4}{10} \times \frac{4}{5} =$



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4) $1\frac{2}{4} \times \frac{5}{6} =$

5) $1\frac{4}{12} \times \frac{2}{12} =$

6) $3\frac{1}{8} \times \frac{1}{4} =$

7) $3\frac{2}{3} \times \frac{2}{8} =$

8) $3\frac{1}{4} \times \frac{2}{6} =$

9) $2\frac{1}{2} \times \frac{2}{6} =$

10) $3\frac{1}{6} \times \frac{1}{2} =$



• Solve the division.

1) $\frac{1}{7} \div \frac{3}{4} =$

2) $\frac{3}{10} \div \frac{5}{8} =$

3) $\frac{6}{11} \div \frac{2}{5} =$

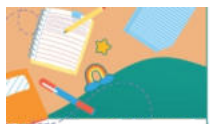
4) $\frac{5}{3} \div \frac{3}{7} =$

5) $\frac{6}{5} \div \frac{7}{9} =$

6) $\frac{3}{2} \div \frac{4}{15} =$

7) $\frac{3}{10} \div \frac{1}{8} =$

8) $\frac{4}{9} \div \frac{7}{3} =$



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9) $\frac{10}{7} \div \frac{3}{4} =$

10) $\frac{8}{5} \div \frac{4}{3} =$